

```

#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;

    printf("Enter size of input: ");
    scanf("%d", &n);

    // O(1) Space Complexity
    int constantVar = 100;
    printf("O(1) Space: %d\n", constantVar);

    // O(n) Space Complexity
    int *linearArray = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        linearArray[i] = i;
    }
    printf("O(n) Space allocated for %d elements.\n", n);

    // O(n^2) Space Complexity
    int **matrix = (int **)malloc(n * sizeof(int *));
    for (int i = 0; i < n; i++) {
        matrix[i] = (int *)malloc(n * sizeof(int));
    }
    printf("O(n^2) Space allocated for %d x %d matrix.\n", n, n);

    free(linearArray);

    for (int i = 0; i < n; i++) {
        free(matrix[i]);
    }
    free(matrix);

    return 0;
}

```